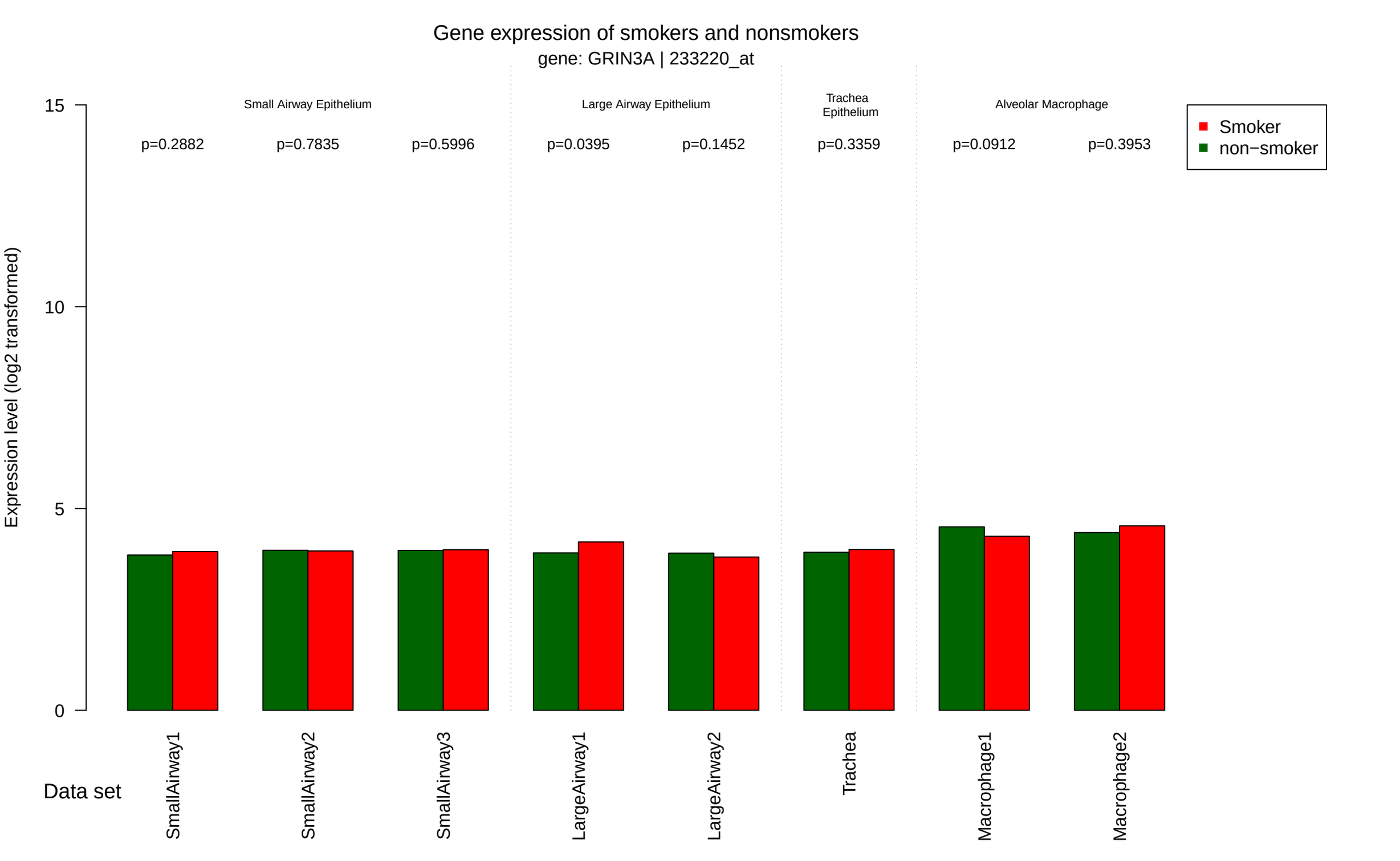
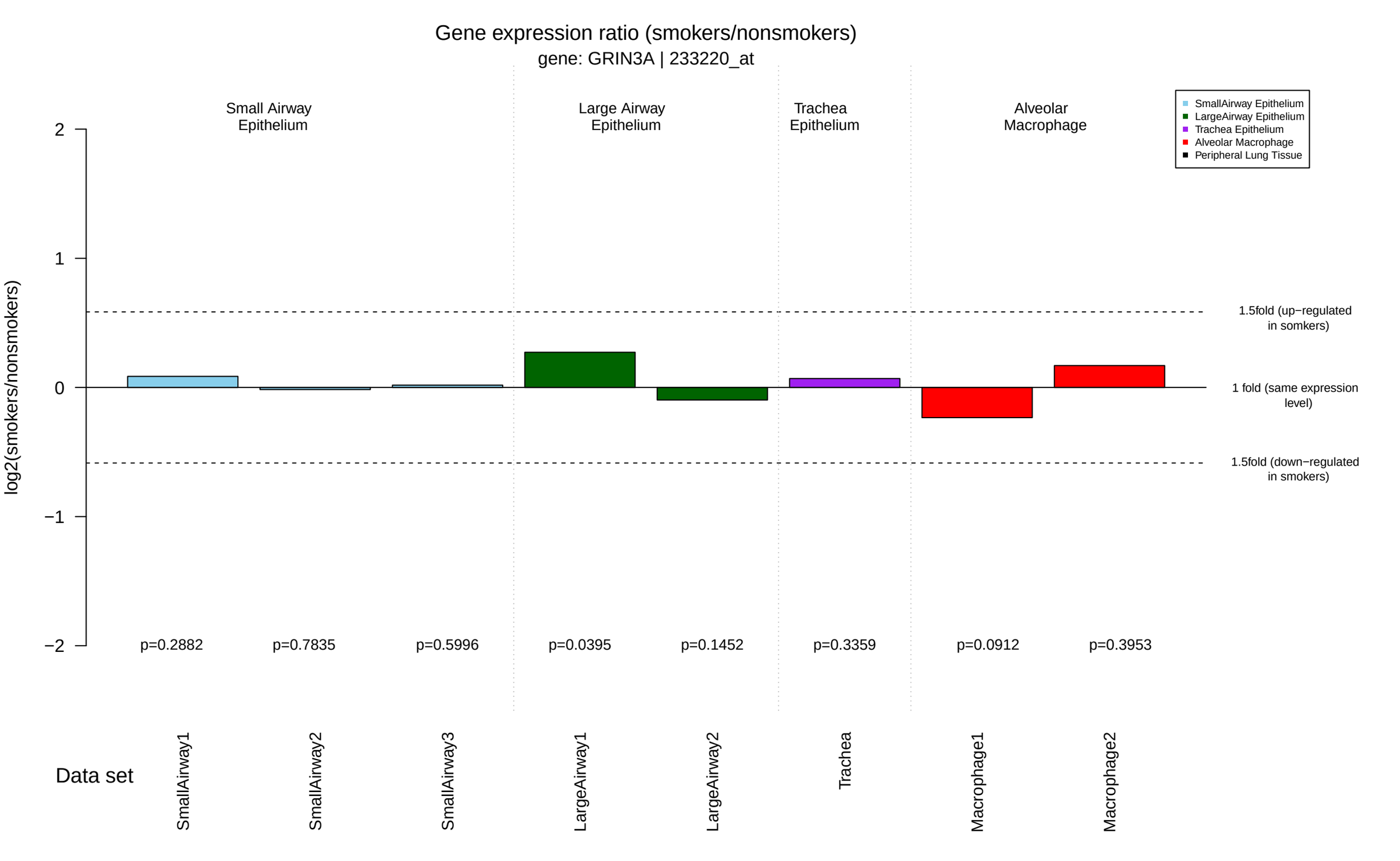
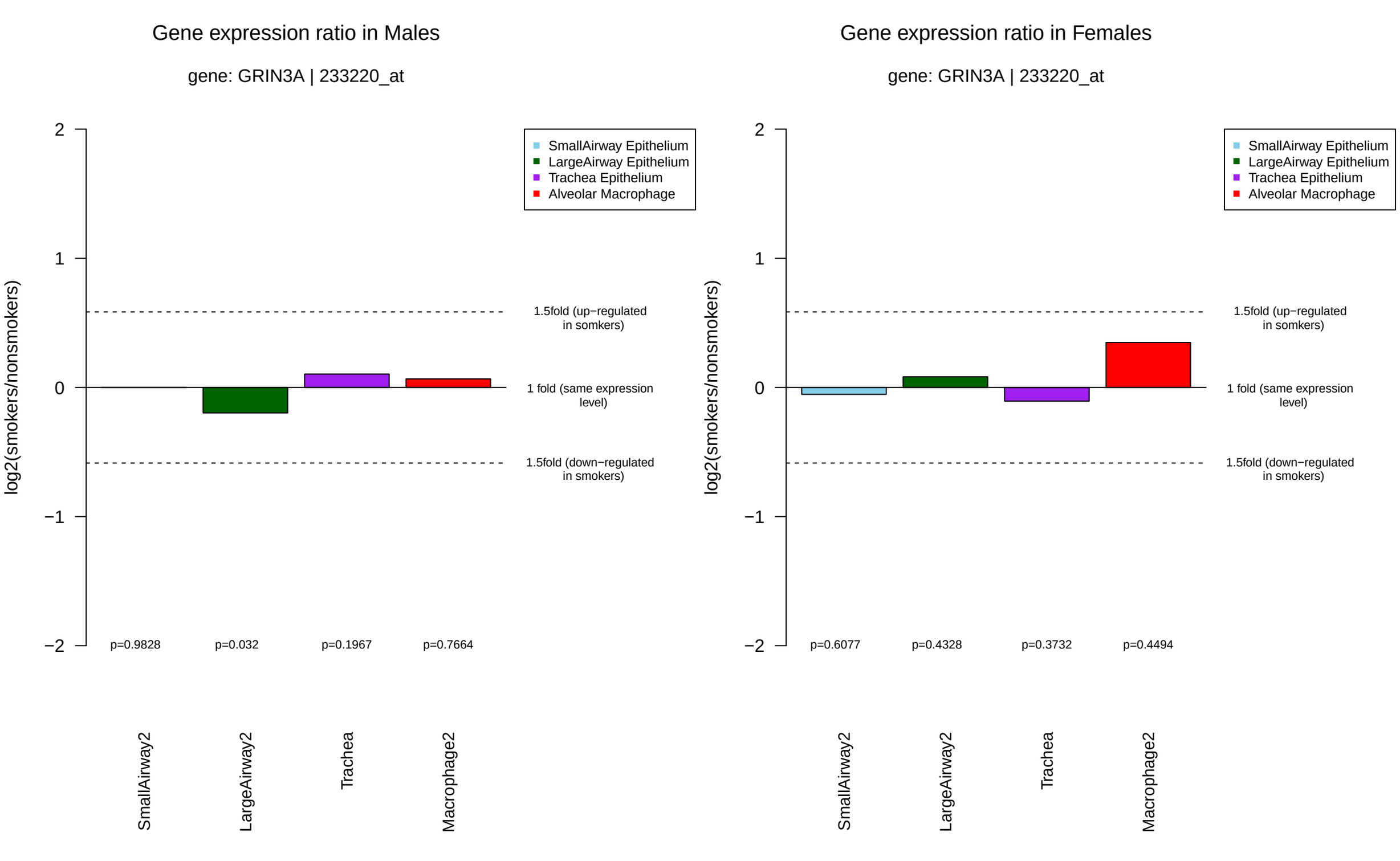
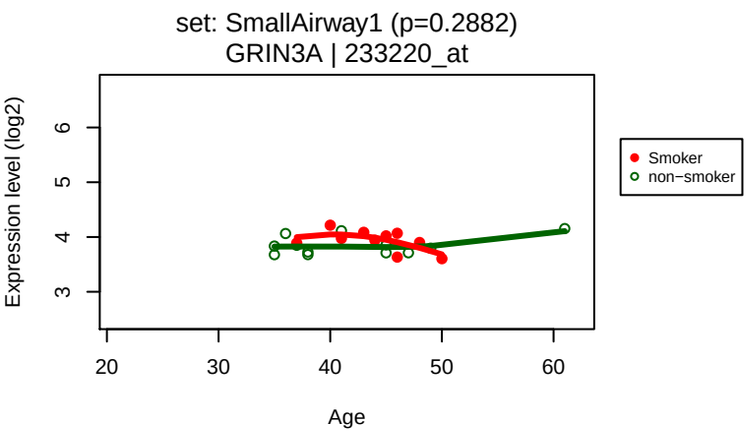
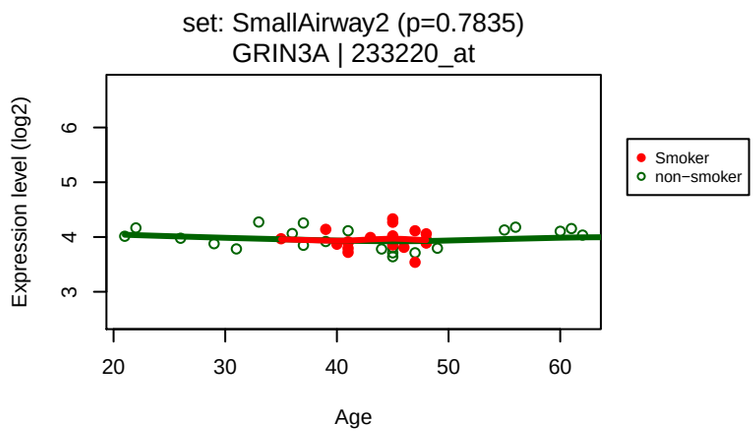
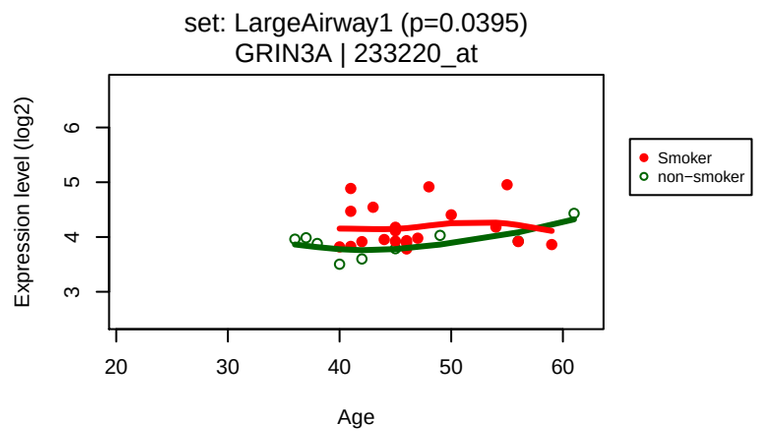

Gene expression of smokers and nonsmokers
gene: GRIN3A | 233220_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2882
p=0.7835
p=0.5996
p=0.0395
p=0.1452
p=0.3359
p=0.0912
p=0.3953
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

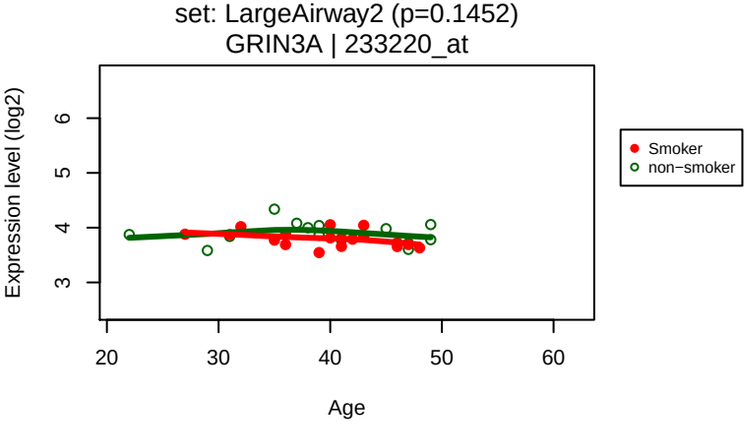

Gene expression ratio (smokers/nonsmokers)
gene: GRIN3A | 233220_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2882
p=0.7835
p=0.5996
p=0.0395
p=0.1452
p=0.3359
p=0.0912
p=0.3953
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

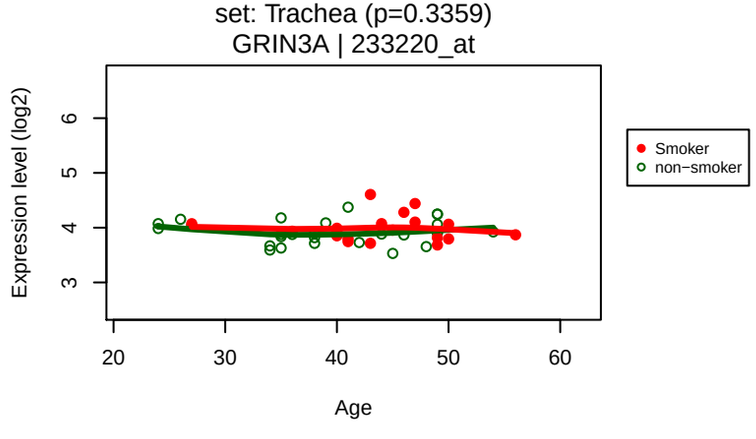

Gene expression ratio in Males
gene: GRIN3A | 233220_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9828
p=0.032
p=0.1967
p=0.7664
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRIN3A | 233220_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6077
p=0.4328
p=0.3732
p=0.4494
SmallAirway2
LargeAirway2
Trachea
Macrophage2

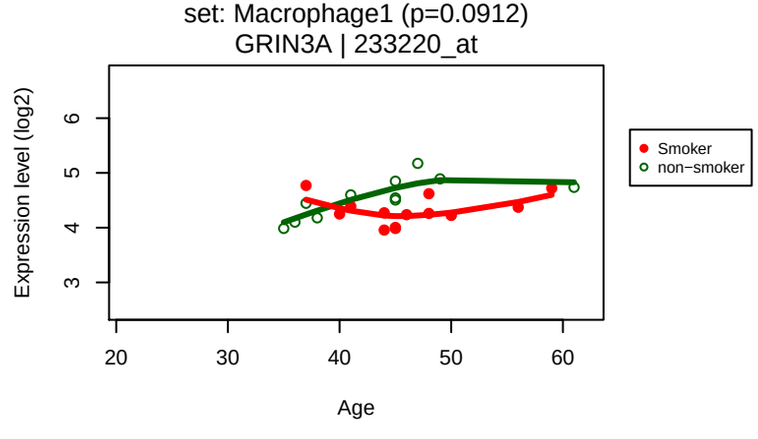

set: SmallAirway1 (p=0.2882)
GRIN3A | 233220_at
Expression level (log2)
Age
Smoker
non−smoker

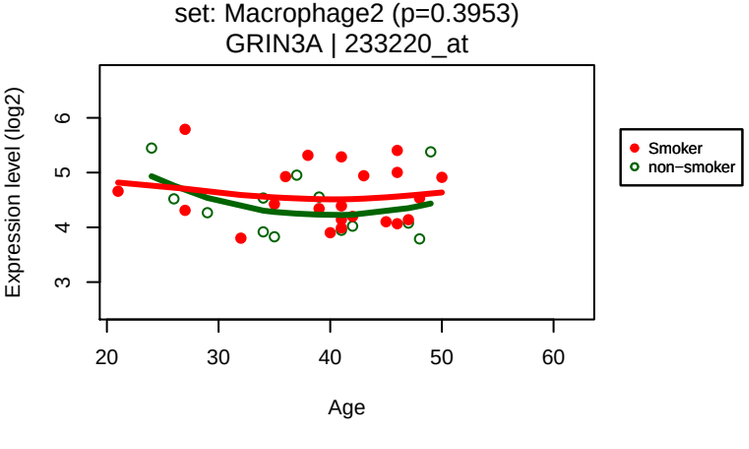

set: SmallAirway2 (p=0.7835)
GRIN3A | 233220_at
Expression level (log2)
Age
Smoker
non−smoker

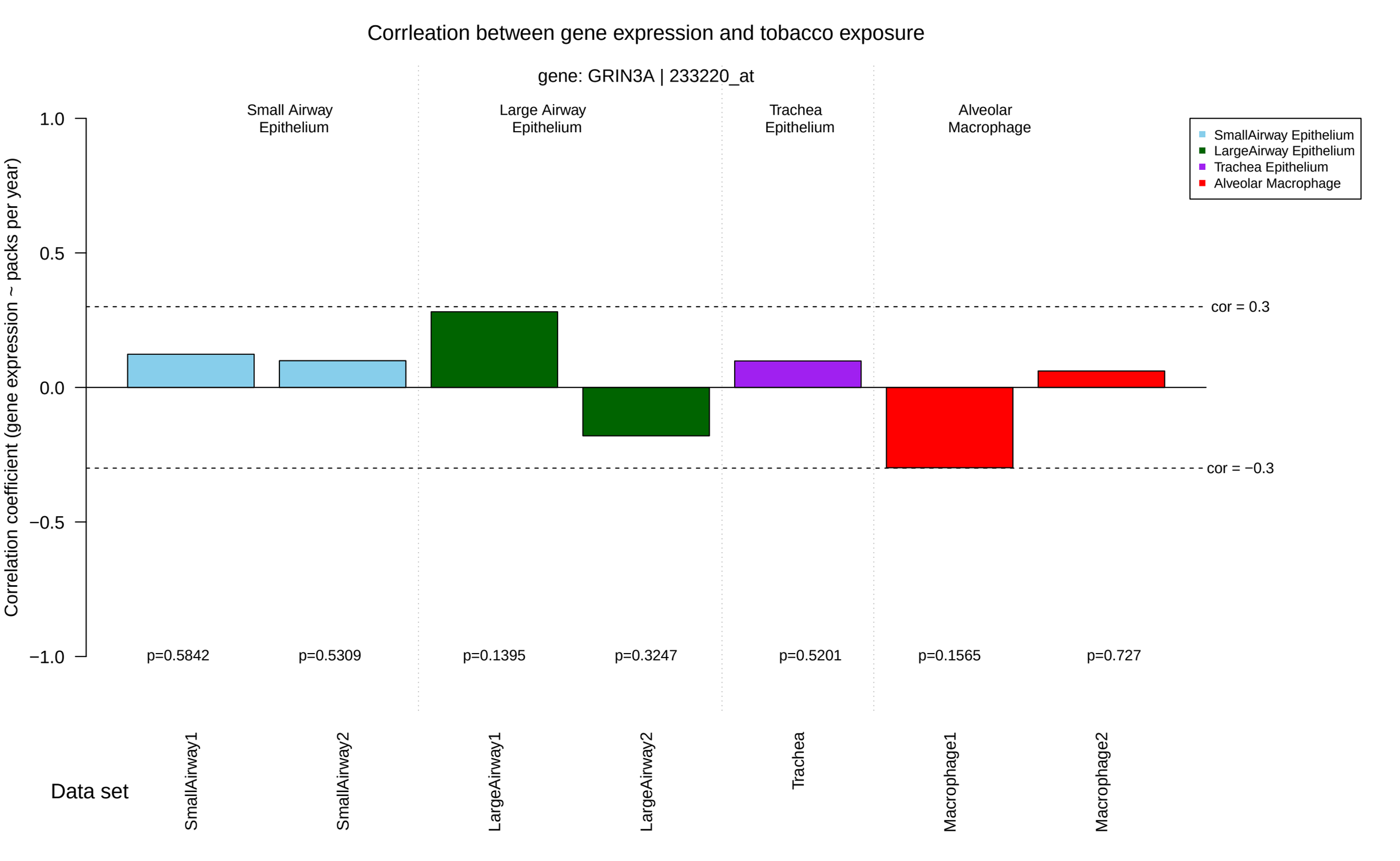

set: LargeAirway1 (p=0.0395)
GRIN3A | 233220_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1452)
GRIN3A | 233220_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3359)
GRIN3A | 233220_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0912)
GRIN3A | 233220_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3953)
GRIN3A | 233220_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRIN3A | 233220_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5842
p=0.5309
p=0.1395
p=0.3247
p=0.5201
p=0.1565
p=0.727
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2